

Module 6

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Module 6

Fundamentals of Inheritance

In Module 6 we introduce the notion of inheritance. The use of inheritance allows us to take a base class object, which can be thought of as a crude object, and from it develop a more refined object, which is known as a derived class object. The derived class object has all the properties of the base class, plus the additional "refinements" specified by the derivation. Inheritance plays an important role in software reuse, i.e., taking something you've already developed and reusing in another application, perhaps in a somewhat different way.

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Learning Objectives

- Protected scope
- Overriding functions
- Initializers in constructors
- Binding times
- Virtual functions
- Virtual destructors and clone functions

Lectures

Presentation

- **[6.1: Comments on Reading Assignments for Module 5](#)**
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-6-dot-1>)
- **[6.2: Hierarchies](#)**
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-6-dot-2>)
- **[6.3: Derived Classes](#)**
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-6-dot-3>)
- **[6.4: Initializers and Order of Constructor Invocation](#)**
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-6-dot-4>)
- **[6.5: Derived Classes and Operator](#)**
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-6-dot-5>)
- **[6.6: Binding Times](#)**
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-6-dot-6>)
- **[6.7: V-tables and V-ptrs for Implementing Virtual Functions](#)**
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-6-dot-7>)
- **[6.8: Virtual Destructors and Cloning](#)**
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-6-dot-8>)

Reading

The reading for this Module is C review. Depending on your C background, you may be able to move through this material quickly.

Required Readings

- *C++ Primer Plus*, Chapter 14

Recommended Readings

- *Thinking in C++ Volume One*, pp. 497–572

Discussion

[Module 6 Discussion](#)

(https://onlinelearning.berkeley.edu/courses/2291660/discussion_topics/14351910)

Assignment

[Module 6: Short Answer 6](#)

(<https://onlinelearning.berkeley.edu/courses/2291660/assignments/33623158>)

[Module 6: Programming Assignment 6](#)

(<https://onlinelearning.berkeley.edu/courses/2291660/assignments/33623156>)

Source Code

In each module of this course, you will be required to submit a programming assignment. For each module, I have prepared sample source code for your reference. Please take a few minutes to download and install all of these programming examples. The files have been compressed into one zip. You will need a zip utility to decompress the file.

 [sources.zip](#)

<https://onlinelearning.berkeley.edu/courses/2291660/files/313890817/download?wrap=1>

Reminder: Your Course End Date

As you work through the course, please keep your course End Date in mind, which is indicated in the initial e-mail you received when you enrolled. If you want to save any commentary or assignments for future reference, please make sure to print or copy/paste those materials before your end date.